

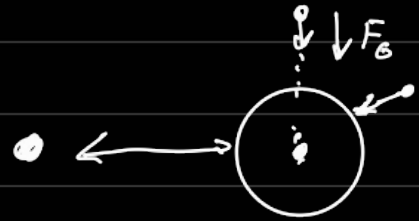
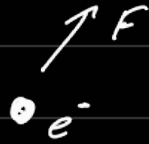
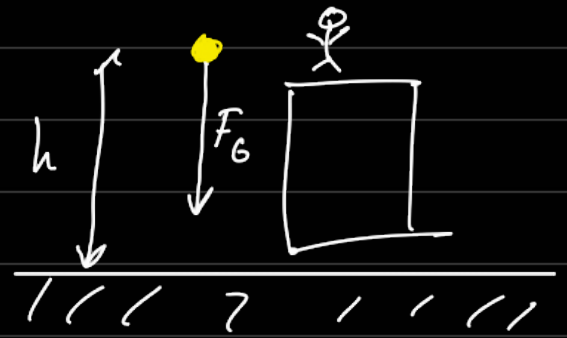
→ Normalized parameters:

$$* m \frac{d^2 z}{dt^2} = -mg *$$

$$F_G = ma$$

\* "g = 1", "m = 1" \*

$$F_G = -mg = -1$$



\* Central forces:

$$* F = -\frac{d}{dx} V(x) *$$

$$• F_G(z) = -\frac{d}{dz} V(z) = -1$$

$$• F_G = -mg = -\frac{d}{dx} V(x) \Rightarrow mg = \frac{dV}{dx} \Rightarrow * V(x) = mgx *$$

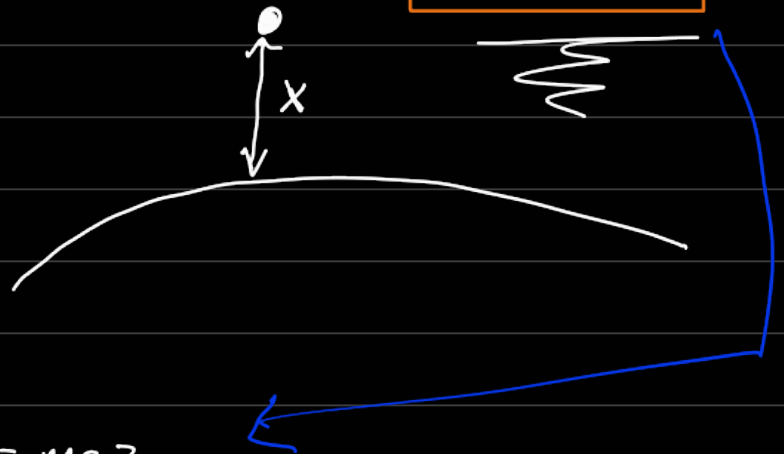
V(x): potential Energy:

$$\frac{dV}{dx} = mg = -F_G$$

Grav Pot.

\* Notation: Same use

- V(x) or
- U(x)



$$V(z) = mgz$$

$$= z \leftarrow$$

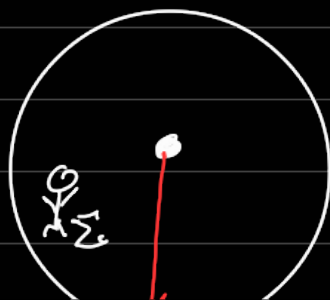
$$m = g = 1$$

$$V(z) = z$$

↳

$$|4| S_a = \int_0^1 (T_{kin}(z'_a) - V(z_a(t))) dE$$

PS4/ ①



- $\vec{r}_{\Sigma,app} = \omega^2 \vec{r}_{\Sigma} + 2\omega (\dot{\xi} \hat{e}_n - \dot{\eta} \hat{e}_\xi)$

→  $\vec{r}_S = (0, -Ut)$

$\vec{V}_S = \frac{d\vec{r}_S}{dt} = (0, -U) \leftarrow \text{constant velocity}$

$\vec{a}_S = \frac{d^2\vec{r}}{dt^2} = \frac{d\vec{V}_S}{dt} = (0, 0)$

\* Now, go to eq. (143 - notes)

• Simply 143:

$$m a_{\Sigma,app} = F - \textcircled{1} - \textcircled{2} - \textcircled{3} - \textcircled{4}$$

$\begin{matrix} \text{X} & \text{X} & \text{X} & \text{---} \end{matrix}$

Any external force

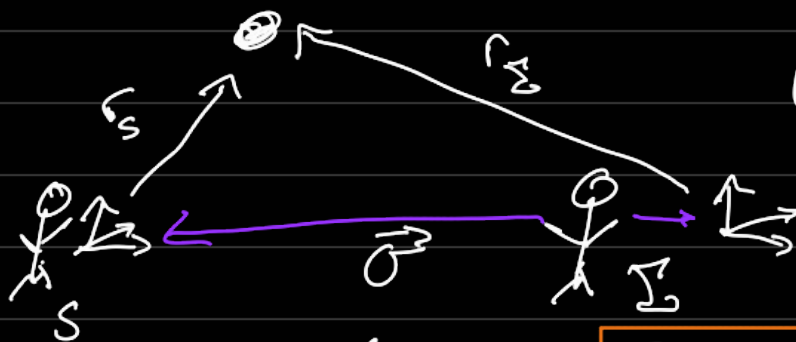
→ Recall  $a_{\Sigma} = \vec{0}$

①  $-m \vec{\sigma}''(t) = 0$

• For us this is zero

why: Because for both people in  $S, \Sigma$  our coord systems have the same origin.

$\vec{\sigma}(t)$



(Pg 37)

same origin → overlap →

$\vec{\sigma} = 0$

② Euler force:  $-m (-\omega \times \vec{r}_{\Sigma}(t)) = 0$

\* Claim: Also zero because  $\frac{d\omega}{dt} = 0$

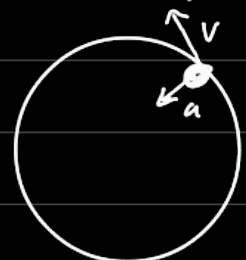
③ / ④: Remain: we'll have to solve for these

→

F<sub>ext</sub>: Also zero: the problem doesn't add any external forces

$\vec{F}_{ext} = 0$

Force from motion:  $m \ddot{r} = F_m$



ex:  $F_{ex}$  could be gravity  $F_g = -mg$

$\vec{L}$  or  $\vec{F}$  you pushing

\* Tech: \*  $\vec{r}_S = (0, -vt)$  \*

(Impulse  $\Rightarrow$  motion etc..)

Back to 143:

$$a_{\Sigma, app} = -2(\vec{\omega} \times \vec{V}_{\Sigma, app}) - M(\vec{\omega} \times (\vec{\omega} \times \vec{r}_S))$$

(a)

(b)

$$\vec{r}_S = (0, -vt)$$

$$\vec{r}_\Sigma = \xi(t) \hat{e}_\xi(t) + \eta(t) \hat{e}_\eta(t)$$

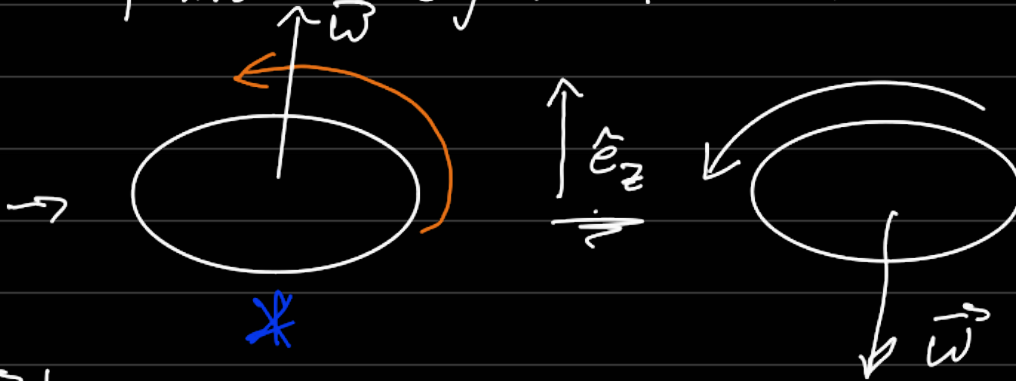
$$\dot{\vec{r}}_{\Sigma, app} = \dot{\xi}(t) \hat{e}_\xi(t) + \dot{\eta}(t) \hat{e}_\eta(t)$$

$$\ddot{\vec{r}}_{\Sigma, app} = \ddot{\xi}(t) \hat{e}_\xi(t) + \ddot{\eta}(t) \hat{e}_\eta(t)$$

$$\dot{\vec{r}}_{\Sigma, app} = \vec{V}_{\Sigma, app}$$

\* Definition:  $\omega = \frac{d\varphi}{dt}$  (angular velocity)

$\vec{\omega}$ : points according to the R.H.-rule



$$|\vec{\omega}| = \omega$$

magnitude of  $\vec{\omega}$  is  $\omega$  (ang. vel.)

• Speed  $V = |\vec{V}|$

$$\Rightarrow \vec{\omega} = \omega \hat{e}_z$$

(a)  $-2(\vec{\omega} \times \vec{V}_{\Sigma, app}) M$  \*

• Can't use  $[\hat{e}_x, \hat{e}_y, \hat{e}_z]$  because both vectors have different basis.

• Only use  $[\hat{e}_\xi, \hat{e}_\eta]$  if both vectors have the same basis

expand:

$$= -2 \left( (\omega \hat{e}_z) \times (\dot{\xi} \hat{e}_z + \dot{\eta} \hat{e}_\eta) \right)$$

$$= -2 \omega \dot{\xi} (\hat{e}_z \times \hat{e}_\eta) - 2 \omega \dot{\eta} (\hat{e}_z \times \hat{e}_z)$$

$$\hookrightarrow \vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\hat{e}_z \times \hat{e}_\eta = \hat{e}_\xi$$

$$\hat{e}_z \times \hat{e}_\eta = -\hat{e}_\eta \times \hat{e}_z = -\hat{e}_\xi$$

Plug in

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$$\textcircled{a} = -2 \omega \dot{\xi} \hat{e}_\eta m + m 2 \omega \dot{\eta} \hat{e}_\xi = -2 \omega (\dot{\xi} \hat{e}_\eta - \dot{\eta} \hat{e}_\xi) m$$

$$\textcircled{b} -m (\vec{\omega} \times (\vec{\omega} \times \vec{r}_\Sigma))$$

Option 1 expand as we did above  $\vec{r}_\Sigma = \begin{pmatrix} \end{pmatrix}$   
 $\vec{\omega} = \omega \hat{e}_z$

Option 2 Identities

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})$$

$$\vec{A} = \vec{\omega}$$

$$\vec{B} = \vec{\omega}$$

$$\vec{C} = \vec{r}_\Sigma$$

$$\vec{\omega} \times (\vec{\omega} \times \vec{r}_\Sigma) = \vec{\omega} (\vec{\omega} \cdot \vec{r}_\Sigma) - \vec{r}_\Sigma (\vec{\omega} \cdot \vec{\omega})$$

$$\vec{\omega} \cdot \vec{r}_\Sigma = 0 \quad \leftarrow \text{Since } \vec{r} \text{ has no } z\text{-component}$$

$$\vec{\omega} = \omega \hat{e}_z$$

$$\vec{\omega} \cdot \vec{\omega} = \omega^2$$

$$-m (\vec{\omega} \times (\vec{\omega} \times \vec{r}_\Sigma)) = -m (-\vec{r}_\Sigma \omega^2)$$

$$= +m \omega^2 \vec{r}_\Sigma$$

$$\textcircled{b} = m \omega^2 \vec{r}_\Sigma$$

$$m a_{\Sigma, \text{app}} = \textcircled{a} + \textcircled{b}$$

$$= m \omega^2 \vec{r}_\Sigma - 2 \omega m (\dot{\xi} \hat{e}_\eta - \dot{\eta} \hat{e}_\xi)$$

$$a_{\Sigma,app} = \ddot{\vec{r}}_{\Sigma,app} = \omega^2 \vec{r}_{\Sigma} - 2\omega(\dot{\xi} \hat{e}_n - \dot{\eta} \hat{e}_{\xi})$$

